REFLEXIVE MODULES ARE TORSION FREE AND THE DOUBLE DUAL MAP IS INJECTIVE EXACTLY FOR FINITE TORSION FREE MODULES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The double dual map j : M to Hom(Hom(M, R), R) and the definition of a reflexive module. definition environment at source lines 5148-5157, immediately before the counted statement. Supplies the map whose kernel, cokernel and injectivity the three assertions of the statement are about.
- The module of maps into a torsion free module is torsion free, applied to the double dual of R. the author cites this at proof lines 5178-5179 as Lemma lemma-hom-into-torsion-free (tag 0AUX), the preceding lemma of the same chapter. Gives assertion (1) immediately and also shows injectivity of j for torsion free M in the last line of the proof.
- Finitely generated modules over a domain have finite rank, and M tensor K is a finite dimensional vector space. used at proof lines 5180-5187 to choose the generators x_1 to x_r that map to a basis and to clear denominators. Produces the partial generating system and the element c with which the two comparison maps alpha and beta are built.

Source reference. Original lemma 0AV0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13668–13668. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AUX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13665–13665. Navigation pointer to the immediately preceding lemma of the same chapter, which the author invokes for the torsion freeness of the double dual; that statement is counted by the sibling card S1-algebra-hom-into-torsion-free of this lane, and the construction with alpha, beta and the diagram remains inside the counted proof span.

Lemma 0.2. *Let R be a domain and let M be an R-module.*

(1) *If M is reflexive, then M is torsion free.*
(2) *If M is finite, then the kernel and cokernel of $j : M \to \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)$ are torsion modules.*
(3) *If M is finite, then j is injective if and only if M is torsion free.*

Proof. To see (1) use that the double dual of M is torsion free by Lemma 0AUX. Assume that M is finite. Choose generators $x_1, \ldots, x_n$ of M over R. Let K be the fraction field of R. After renumbering we may assume that for some $0 \leq r \leq n$ the elements $x_1, \ldots, x_r$ map to a basis of $M \otimes_R K$ over K. For $r < i \leq n$ write $x_i \otimes 1 = \sum_{1 \leq j \leq r} x_j \otimes k_{ij}$ with $k_{ij} \in K$. Choose $c \in R$ such that $a_{ij} = ck_{ij} \in R$ and such that $cx_i = \sum_{1 \leq j \leq r} a_{ij} x_j$ in M (small detail omitted). Then we get maps

$$R^{\oplus r} \xrightarrow{\alpha} M \xrightarrow{\beta} R^{\oplus r} \xrightarrow{\alpha} M$$

with both $\alpha \circ \beta$ and $\beta \circ \alpha$ given by multiplication with c. Here α uses $x_1, \ldots, x_r$ and β sends x_i to c times the ith basis vector for $1 \leq i \leq r$ and to the vector $(a_{i1}, \ldots, a_{ir})$ for $r < i \leq n$. Consider the commutative diagram

$$\begin{array}{ccccccc}
R^{\oplus r} & \longrightarrow & M & \longrightarrow & R^{\oplus r} & \longrightarrow & M \\
\downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle j} \\
R^{\oplus r} & \longrightarrow & \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R) & \longrightarrow & R^{\oplus r} & \longrightarrow & \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)
\end{array}$$

Of course the compositions of consecutive arrows on the bottom row are equal to multiplication by c. The diagram shows that the kernel and cokernel of j are annihilated by c which proves (2). For (3) note that if M is torsion free, then β is injective which implies that j is injective. The converse is clear because we've seen already that the double dual of M is torsion free. $\square$